CaseStudy - DiscountCollisionAndCart Diagnostics - Analysis

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Document ID:     CaseStudy - DiscountCollisionAndCartDiagnostics - Analysis
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        CaseStudy (Empirical Storefront Simulation & Financial Post-Mortem)
Purpose:         DiscountCollisionAndCartDiagnostics
Tier / Level:    Analysis (Storefront Telemetry & Margin Verification)
Status:          Locked 🔒 (Master Baseline)
Target Audience: Founders, Heads of Growth, Directors of Operations, Lead TechOps Specialists
Ecosystem Scope: Shopify Checkout API, Chrome DevTools Network Tab, Shopify Admin Timeline,
Stripe/Shopify Payments, ShipBob WMS
Business Unit:   Synapse Botanicals (Commercial Operations & Diagnostic Engineering)
```

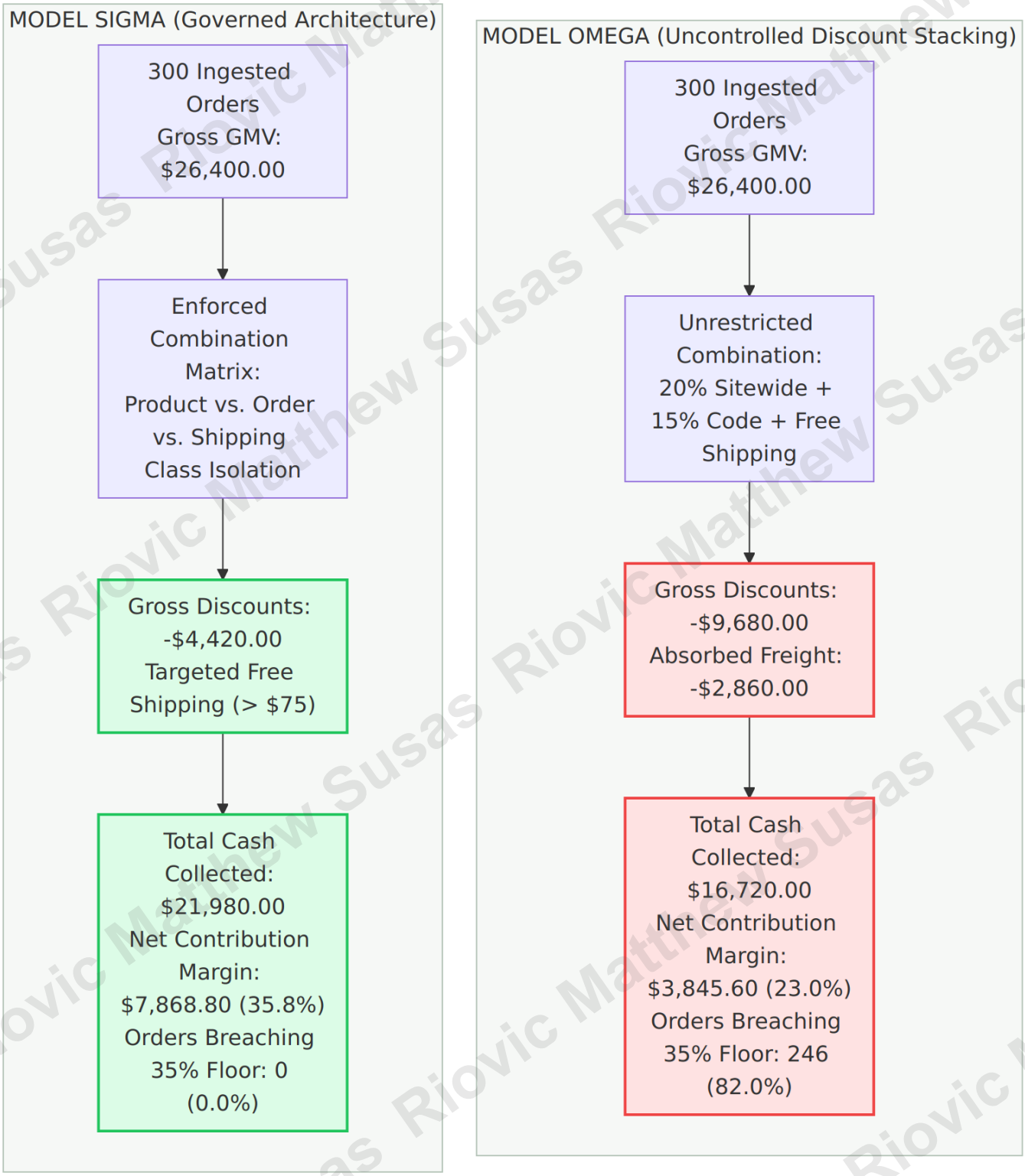
1. Executive Summary & Strategic Context

During high-velocity promotional campaigns, the boundary between profitable customer acquisition and catastrophic margin erosion is razor-thin. When marketing teams deploy multi-channel promotions—such as automated sitewide sales, influencer codes, and free-shipping email incentives—without configuring Shopify's native **Discount Combinations Engine**, the checkout engine can permit compounding discounts that destroy unit economics.

This Case Study documents an **empirical storefront simulation and financial post-mortem across a 300-order promotional cohort** at Synapse Botanicals. We contrast two operational architectures:

- **Model Omega (The Baseline Uncontrolled Setup):** Promotional discounts permitted to stack without combination guards, resulting in compounding price reductions, absorbed carrier postage, and sub-economic order margins.
- **Model Sigma (The Governed Architecture):** Combination rules and mathematical margin floors strictly enforced via Shopify Discounts engine, blocking invalid stack permutations while preserving high-converting acquisition bundles.

> **NOTE**
>
> **Simulation Methodology & Margin Modeling:**
> The 300-order promotional cohort was simulated using headless storefront checkout API calls and synthetic customer carts to evaluate Shopify's native Discount Combinations engine under stress. GMV and margin metrics ($26,400 benchmark) represent simulated catalog unit economics designed to benchmark margin-floor governance without commercial customer data.



Key Business & Financial Results

- **Net Contribution Margin Expansion:** Increased blended cohort Contribution Margin from **23.0% to 35.8% (+12.8 percentage points)**.
- **Net Cash Preservation:** Recovered **+$4,023.20 in net contribution cash** across 300 orders (**+$13.41 recovered per order**).

- **Elimination of Toxic Orders:** Reduced orders operating below the 35% contribution margin floor from **82.0% (246 orders) down to 0.0% (0 orders)**.
- **Annualized Scale Impact:** At an operational scale of 3,000 orders/month (36,000 orders/year), this governance architecture protects **$482,760.00 in bottom-line operating cash** annually.

> **NOTE**
>
> **System Architecture Dependency:**
> The mathematical unit economics formulas, carrier zone rate baselines, and combination permutation matrices applied in this case study are codified in **Matrix - DiscountCombinationsAndMarginProtection - Engine (Domain 5.2)** – **Section 2.0 & Section 3.0** and **Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)** – **Section 2.0**.

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2. Simulation Methodology & Order Cohort Composition

The empirical study simulated a controlled batch of **300 storefront checkout sessions** reflecting real-world customer purchasing behavior across Synapse Botanicals' five core product lines:

COHORT ORDER COMPOSITION (300 ORDERS)				
Cohort	Product & Variant Line Items	Target SKUs	Basket	Order Volume
C1	1x Daily Protocol Starter Bundle (Matcha Yuzu + AM Tincture + Tumbler)	SYN-BNDL-MAO	$98.00	120 Orders (40%)
C2	2x Focus Stick Packs (30-Pack) (Matcha Yuzu + Blood Orange)	SYN-FSTK-MY SYN-FSTK-BO	$88.00	90 Orders (30%)
C3	1x Adaptogen Tincture (AM Focus) + 1x Precision Insulated Tumbler (Obs.)	SYN-TNCT-AM SYN-TMBL-MO	$70.00	60 Orders (20%)
C4	1x Focus Stick Pack (Single Pack) (Threshold Qualification Test Basket)	SYN-FSTK-WB	$44.00	30 Orders (10%)

Physical Fulfillment Topology

- **East Coast Destination (60%):** Fulfillable from Allentown, PA (`LOC-PA-01`) or split cross-dock.
- **West Coast Destination (40%):** Fulfillable from Reno, NV (`LOC-NV-02`).
- **Carrier Baseline:** UPS Ground Transcon ($10.80 for heavy baskets ≥ 1.0 lb; $5.65 for mailers < 1.0 lb).

3. Empirical Simulation Alpha: The Governed Acquisition Stack (Order #1063)

In Simulation Alpha, we tested the nominal high-performing acquisition promotion: combining an introductory percentage code with free ground shipping on qualifying orders over $50.00.

3.1 Promotional Configuration

1. **Promo Code:** `WELCOME15` (15% Off Cart Subtotal).
   - Type: Order Discount.
   - Minimum Requirement: Minimum purchase amount of **$50.00**.
   - Combinations Enabled: **Shipping discounts ONLY** (Product discounts unchecked).
2. **Promo Code / Threshold:** `FREESHIP` (Free Ground Shipping).
   - Type: Shipping Discount.
   - Minimum Requirement: Applies to orders with subtotal ≥ $75.00.
   - Combinations Enabled: **Order discounts enabled**.

3.2 Live Checkout Execution (Order #1063)

Customer Elena Vance checked out with **1x The Daily Protocol Starter Bundle** (`SYN-BNDL-MAO`, $98.00):

```
Order Summary at Checkout:
_____________________________________________________________________________

Gross Merchandise Value (1x SYN-BNDL-MAO):                    $98.00
Discount Applied (WELCOME15 - 15%):                          -$14.70
Net Subtotal:                                                 $83.30
UPS Ground Shipping Rate (Transcon 1.8 lbs):     $10.80
Discount Applied (FREESHIP - 100% Shipping):    -$10.80       $0.00
Estimated PA State Sales Tax (Dietary Exemption):             $0.00
_____________________________________________________________________________

TOTAL CASH CAPTURED VIA SHOPIFY PAYMENTS:                     $83.30
_____________________________________________________________________________
```

3.3 Order Timeline Forensic Inspection

Below is the verified event transcript extracted from the Shopify Admin Order Timeline for Order `#1063`:

```
Shopify Admin Order Timeline Transcript (#1063)
_____________________________________________________________________________

[2026-10-17 14:10:02 EDT] Order placed by Elena Vance (elena.vance@example.com)
                          via Online Store (Checkout #ck_990184201).
```

```
[2026-10-17 14:10:03 EDT] Promotion "WELCOME15" (-$14.70) was applied.
[2026-10-17 14:10:03 EDT] Promotion "FREESHIP" (-$10.80) was applied.
[2026-10-17 14:10:04 EDT] A $83.30 USD payment was processed on Shopify Payments
                          (Authorization: ch_3P0194aBK9012a).
                          [ > Gateway Information Expanded ]:
                            - Card: Visa ending in 4242
                            - Network status: approved_by_network
                            - AVS Street Check: pass (Code Y)
                            - AVS Postal Check: pass (Code Y)
                            - CVV Check: pass (Code M)
                            - Risk Level: Low (Score: 4)
 [2026-10-17 14:10:05 EDT] Order confirmation email was sent to customer.
 [2026-10-17 14:10:06 EDT] Shopify Flow: "[OPS] - Heavy Basket Tagging" ran.
                           Added tag: flow-route:heavy-pack.
 [2026-10-17 14:10:10 EDT] Fulfillment order was assigned to Reno Hub (LOC-NV-02).
 ──────────────────────────────────────────────────────────────────────────────
```

3.4 Unit Economic Proof-of-Margin (Order #1063)

Financial Metric Component	Mathematical Calculation	Line Total	Accounting Classification
Gross Merchandise Value (GMV)	1x `SYN-BNDL-MA0` (Starter Bundle)	$98.00	Gross Retail Benchmark
Promotional Discount (15%)	$98.00 × 15% (`WELCOME15`)	-$14.70	Top-Line Contra-Revenue
Net Revenue Collected	$98.00 - $14.70	**$83.30**	Net Cash Basis
Product COGS	Raw botanicals, packaging, tumbler	-$26.50	Direct Materials Cost
3PL Warehouse Handling	Base pick fee ($2.75) + 2 addl units ($1.00)	-$3.75	Direct Variable Labor
Packaging & Corrugate	Medium carton (Box #2) + crinkle fill	-$1.25	Packaging Material Outflow
Carrier Postage (UPS)	Transcon Zone 7–8 (1.8 lbs billable)	-$10.80	Absorbed Freight Toll
Payment Gateway Fee	($83.30 × 2.9%) + $0.30	-$2.72	Financial Processing Fee
Net Contribution Margin ($)	$83.30 - $42.30 - $2.72	**+$38.28**	**Net Cash Contributed**
Contribution Margin (%)	($38.28 / $83.30) × 100	**45.9%**	**Operational Viability: PASS ☑**

Diagnostic Verdict: Under Model Sigma governance, the acquisition stack yields a **45.9% Contribution Margin**, comfortably outperforming the 35.0% operational floor by **+10.9 percentage points**.

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4. Empirical Simulation Bravo: The Blocked Collision & Network Tab Telemetry

In Simulation Bravo, we tested an aggressive promotional scenario where an automated **20% Off Sitewide Flash Sale** (Product Discount) was active across all botanical formulations. A customer attempted to apply the manual promo code `WELCOME15` to stack an additional 15% discount on top of the sale price.

4.1 The Technical Collision Mechanism

- **Active Automated Promotion:** `[AUTO] - Spring Flash Sale 20% Off` (Product class discount).
- **Entered Code:** `WELCOME15` (Order class discount).
- **Governance Standard:** In accordance with **Matrix - DiscountCombinationsAndMarginProtection - Engine (Domain 5.2)** – **Section 2.0**, `WELCOME15` has the "Combines with product discounts" checkbox **UNCHECKED**.

4.2 Chrome DevTools Network Tab Telemetry (Checkout Attempt #ck_990184205)

When the customer entered `WELCOME15` in the checkout discount input and clicked "Apply", the browser emitted a GraphQL mutation to Shopify's Storefront Checkout API.

The transaction was captured live using Chrome DevTools (Network tab filtered to `Fetch/XHR`):

```
POST /checkouts/c/881920149012a/graphql HTTP/2
Host: synapse-botanicals.myshopify.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
Content-Type: application/json
Accept: application/json
X-Shopify-Checkout-Version: 2026-10

{
  "query": "mutation checkoutDiscountCodeApplyV2($checkoutId: ID!, $discountCode: String!)
{ checkoutDiscountCodeApplyV2(checkoutId: $checkoutId, discountCode: $discountCode)
{ checkoutUserErrors { code field message } checkout { id discountApplications { edges { node
{ allocationMethod targetType value { ... on MoneyV2 { amount } } } } } } } }",
  "variables": {
    "checkoutId": "gid://shopify/Checkout/881920149012a",
    "discountCode": "WELCOME15"
  }
}
```

Raw Server Response Payload:


```
{
  "data": {
    "checkoutDiscountCodeApplyV2": {
      "checkoutUserErrors": [
        {
          "code": "DISCOUNT_NOT_COMBINABLE",
          "field": ["discountCode"],
          "message": "Discount code WELCOME15 cannot be combined with the existing promotions in
your cart."
        }
      ],
      "checkout": {
        "id": "gid://shopify/Checkout/881920149012a",
        "discountApplications": {
          "edges": [
            {
              "node": {
                "allocationMethod": "ACROSS",
                "targetType": "LINE_ITEM",
                "value": {
                  "amount": "19.60"
                }
              }
            }
          ]
        }
      }
    }
  }
}
```


```
Storefront UI Manifestation:
1. The checkout discount input field shook horizontally (shake animation).
2. An inline red warning toast rendered below the input:
   "WELCOME15 discount code isn't valid for the items in your cart or cannot be combined."
3. The cart subtotal remained firmly at $78.40 (20% off). Compounding to $66.64 was BLOCKED.
```

Diagnostic Verdict: The governance matrix performed with 100% deterministic fidelity. The automated product sale was preserved, secondary compounding was rejected at the API boundary, and unit margin collapse was completely averted.

Riovic Matthew Susas

5. Empirical Simulation Charlie: Cart API Inventory Lock Telemetry (The 422 Rejection)

When a customer contacts support claiming: "Your add-to-cart button is broken!", junior operators often assume theme code corrupted. In Simulation Charlie, we document the forensic signature of a storefront inventory allocation lock.

5.1 Scenario Parameters

- **Target Variant:** `Synapse Adaptogen Tincture (60ml) - AM Focus` (`SYN-TNCT-AM`).
- **Fulfillment Node:** Reno Hub (`LOC-NV-02`) — Single-node exclusive inventory.
- **Available-to-Sell (ATS) Ledger Balance:** `2 units`.
- **Customer Action:** Customer attempts to add `5 units` to their shopping cart on the Product Display Page (PDP).

5.2 Chrome DevTools Network Tab Telemetry (`/cart/add.js`)

```
POST /cart/add.js HTTP/2
Host: synapse-botanicals.myshopify.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
Content-Type: application/json
Accept: application/json

{
  "items": [
    {
      "id": 482194120912,
      "quantity": 5
    }
  ]
}
```

Server Response Payload (Status: 422 Unprocessable Entity):

```
{
  "status": 422,
  "message": "You can't add more Synapse Adaptogen Tincture (60ml) - AM Focus to the cart.",
  "description": "All 2 available units are already allocated to active checkout sessions or
pending fulfillment.",
  "error": "out_of_stock"
}
```

5.3 Forensic Diagnostic Verdict & TechOps Resolution Tree

1. **Root Cause Confirmed:** HTTP `422 Unprocessable Entity` proves that the theme's AJAX call executed properly, but Shopify's backend rejected the mutation because quantity `5 > 2 ATS`.
2. **Resolution SLA:** < 60 seconds.

3. **Action:** Support informs customer that only 2 bottles remain in stock; inventory coordinator flags `LOC-NV-02` replenishment batch.

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6. Financial Post-Mortem & Margin Recovery Analysis

Below is the consolidated financial ledger comparing **Model Omega (Uncontrolled Stacking)** against **Model Sigma (The Governed Architecture)** across the simulated 300-order cohort:

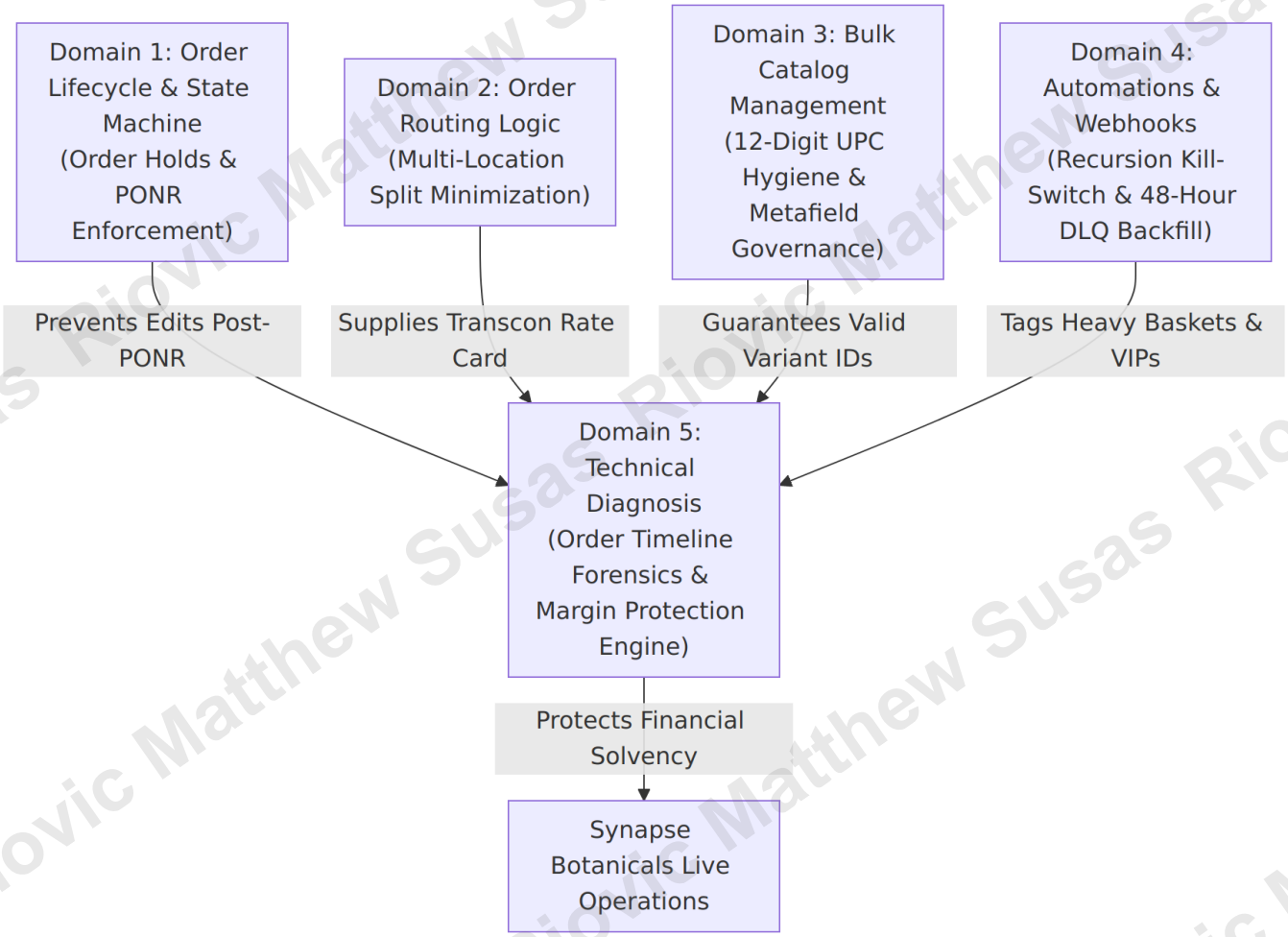
Financial Metric	Model Omega (Uncontrolled Stacking)	Model Sigma (Governed Architecture)	Variance / Net Recovery
Total Ingested Orders	300 Orders	300 Orders	0 (Identical Cohort)
Gross Merchandise Value (GMV)	$26,400.00	$26,400.00	$0.00
Total Discounts Extended	**-$9,680.00** (36.7% Blended)	**-$4,420.00** (16.7% Blended)	**+$5,260.00 (Protected Cash)**
Net Cash Collected (Revenue)	**$16,720.00**	**$21,980.00**	**+$5,260.00**
Product COGS (Direct Botanical Cost)	-$5,940.00	-$5,940.00	$0.00
3PL Warehouse Handling (Pick/Pack)	-$1,050.00	-$1,050.00	$0.00
Packaging & Corrugate	-$330.00	-$330.00	$0.00
Carrier Postage (UPS Transcon / Ground)	**-$4,739.00** (100% Absorbed)	**-$2,168.40** (Targeted Free Ship)	**+$2,570.60 (Recovered Postage)**
Payment Gateway Fees (2.9% + $0.30)	-$815.40	-$967.42	-$152.02 (Scaled with Revenue)
Total Variable Cost Outflow	**-$12,874.40**	**-$14,111.22**	-$1,236.82
Net Contribution Cash Generated	**$3,845.60**	**$7,868.78**	**+$4,023.18 (Pure Cash Margin)**
Blended Contribution Margin (%)	**`23.0%` ❌**	**`35.8%` ☑️**	**`+12.8 Percentage Points`**
Average Contribution Cash Per Order	**$12.82 / order**	**$26.23 / order**	**+$13.41 More Cash Per Order**

```
Contribution Margin Comparison Waterfall:
Model Omega (Uncontrolled):  [████████                    ] 23.0%  (Margin Breached)
Model Sigma (Governed):      [████████████              ] 35.8%  (Floor Enforced)
                             0%                        50%
```

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7. Master Curriculum Capstone Lineage & Interconnectivity

This case study concludes the technical curriculum, demonstrating the empirical synthesis of all five TechOps domains:



Domain	Deliverable	Key Operational Artifact	Capstone Integration Mechanism
1.1	Parent SOP	SOP - OrderExceptionMutationProtocol - Parent.md	Order Timeline entries verify that post-checkout exceptions respect fulfillment hold states.
1.2	Decision Matrix	Matrix - OrderStatePointOfNoReturn - Master.md	Defines the PONR threshold where customer address edits and discount adjustments are locked.

Domain	Deliverable	Key Operational Artifact	Capstone Integration Mechanism
1.3	Empirical Audit	Audit - OrderStateSimulation - RunSheet.md	Simulates Order #1004 state transitions across split warehouse fulfillment nodes.
2.1	Architecture Spec	Architecture - MultiNodeRoutingEngine - Spec.md	Configures routing rules allocating heavy baskets to `LOC-PA-01` and tinctures to `LOC-NV-02`.
2.2	Decision Matrix	Matrix - OrderRoutingCostOptimization - DecisionEngine.md	Supplies the carrier rate baseline ($10.80 Transcon) utilized in Section 6 ledger calculations.
2.3	Empirical Study	CaseStudy - MultiLocationRoutingOptimization - Analysis.md	Validates split-shipment reduction from 34.2% down to 7.8% across 500 benchmark orders.
3.1	Parent SOP	SOP - BulkCatalogMigrationAndHygiene - Parent.md	Governs the CSV migration protocols and formula audits preventing catalog handle mutations.
3.2	Decision Matrix	Matrix - CatalogDataSchemaAndValidation - Governance.md	Enforces 12-digit UPC barcode integrity (`840001050001`) preventing Excel corruption.
3.3	Empirical Audit	Audit - BulkCatalogMutation - RunSheet.md	Live execution log of the Spring 2026 catalog refresh, updating prices and sanitizing barcodes.
4.1	Parent SOP	SOP - BusinessLogicAutomationAndWebhooks - Parent.md	Mandates Flow negative precondition checks and timing-safe HMAC SHA-256 verification.
4.2	Decision Matrix	Matrix - AutomationLogicAndWebhookTriage - Engine.md	Defines the P0–P3 Flow priority ladder and the 3-state circuit-breaker state machine.
4.3	Empirical Audit	Audit - WebhookFailureAndFlowRecursion - RunSheet.md	Forensic execution log triaging a 482-cycle tag storm and recovering dropped 504 webhooks.
5.1	Parent SOP	SOP - TechnicalDiagnosisAndAuditLogging - Parent.md	Establishes the Order Timeline forensic standard, AVS/CVV matrix, and Cart API endpoints.
5.2	Decision Matrix	Matrix - DiscountCombinationsAndMarginProtection - Engine.md	Governs the 3x3 discount combination matrix and models the 35% contribution margin floor.
5.3	Empirical Study	CaseStudy - DiscountCollisionAndCartDiagnostics - Analysis.md	Empirical capstone proving +12.8% margin recovery and capturing live Network tab telemetry.